

IsoSpec2: Ultrafast Fine Structure Calculator, Supporting Information

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Contents

1	Probabilities and masses	S1
2	Other isotopic calculators	S2
3	Order of calculations	S8
4	Subisotopologues	S8
5	Speed-up due to formula order	S9
6	Number of isotopologues	S12
7	Real Punnett tiles	S13
8	Simulations	S14
9	Complexity	S14
10	Computer specs	S15
11	How to run IsoSpec2?	S15
	References	S15

1 Probabilities and masses

In the manuscript, we have described how the calculations proceed on the example of $C_{100}O_6$. For clarity, we show here how this generalizes to a general input for proteomics, which mostly consists of $C_cH_hO_oN_nS_s$, where $c, h, \dots$ represent counts of atoms, all isotopologues. This molecule comprises 5 submolecules: C_c , H_h , N_n , O_o , and S_s . Furthermore, any isotopologue can be written down as a product of constituent subisotopologues as

$$^{12}C_{c_0} \ ^{13}C_{c_1} \ ^1H_{h_0} \ ^2H_{h_1} \ ^{14}N_{n_0} \ ^{15}N_{n_1} \ ^{16}O_{o_0} \ ^{17}O_{o_1} \ ^{18}O_{o_2} \ ^{32}S_{s_0} \ ^{33}S_{s_1} \ ^{34}S_{s_2} \ ^{36}S_{s_4}, \quad (1)$$

where the subscript symbols ($c_0, c_1, \dots$) represent counts of atoms at a given isotopic variant. For the above formulas $c = c_0 + c_1$, $h = h_0 + h_1$, and so on.

We will reserve upper case letters P and M for isotopologue probability and mass. In small-caps we denote probability (natural frequency) and mass of an element as p and m respectively.

The basic models¹ assumes independence of

isotopic variants and element specific frequencies of isotopes. Therefore, probability P of one isotopologue equals a product of subisotopologue probabilities obtained for each submolecule separately,

$$P = \binom{c}{c_0, c_1} p_{12C}^{c_0} p_{13C}^{c_1} \binom{h}{h_0, h_1} p_{1H}^{h_0} p_{2H}^{h_1} \times \binom{n}{n_0, n_1} p_{14N}^{n_0} p_{15N}^{n_1} \binom{o}{o_0, o_1, o_2} p_{16O}^{o_0} p_{17O}^{o_1} p_{18O}^{o_2} \times \binom{s}{s_0, s_1, s_2, s_3} p_{32S}^{s_0} p_{33S}^{s_1} p_{34S}^{s_2} p_{36S}^{s_3}. \quad (2)$$

For instance, subisotopologue of O_o with isotope counts (o_0, o_1, o_2) has probability equal to $\binom{o}{o_0, o_1, o_2} p_{16O}^{o_0} p_{17O}^{o_1} p_{18O}^{o_2}$. Have we discerned positions of isotopes within a molecule, the probability would be equal only to $p_{16O}^{o_0} p_{17O}^{o_1} p_{18O}^{o_2}$, but mass spectrometer cannot do that, as it measures the combined mass of a molecule. Term $\binom{o}{o_0, o_1, o_2}$ counts the different ways these three counts can be distributed among the o atoms.

The mass of an isotopologue M can be calculated by summing masses of subisotopologues of individual submolecules. The masses of individual isotopes, denoted in small-caps as $m^{12C}, m^{13C}, \dots$, are provided by IUPAC.² In mathematical terms,

$$M = m_{12C}c_0 + m_{13C}c_1 + m_{1H}h_0 + m_{2H}h_1 + m_{14N}n_0 + m_{15N}n_1 + m_{16O}o_0 + m_{17O}o_1 + m_{18O}o_2 + m_{32S}s_0 + m_{33S}s_1 + m_{34S}s_2 + m_{36S}s_3. \quad (3)$$

This can be represented in an equivalent expression with the focus on the monoisotopic (lightest) mass and mass differences,

$$M = M_{\text{mono}} + \Delta m_{13C}c_1 + \Delta m_{2H}h_1 + \Delta m_{15N}n_1 + \Delta m_{17O}o_1 + \Delta m_{18O}o_2 + \Delta m_{33S}s_1 + \Delta m_{34S}s_2 + \Delta m_{36S}s_3, \quad (4)$$

where

$$M_{\text{mono}} = m_{12C}c + m_{1H}h + m_{14N}n + m_{16O}o + m_{32S}s \quad (5)$$

denotes the monoisotopic mass, and the differences are calculated with respect to the lightest mass; for example $\Delta m_{13C} = m_{13C} - m_{12C}$ and $\Delta m_{36S} = m_{36S} - m_{32S}$. Note, that M_{mono} is a mass of an actual isotopologue - the lightest one, so that we use upper case to denote its mass.

2 Other isotopic calculators

Here, we show how **IsoSpec2** relates to the other isotope calculators. Namely, we compare to 1D Fourier Transform based **Mercury2**,^{3,4} combinatorial-based **BRAIN**,⁵⁻⁷ and Poisson-approximation based algorithm of Sadygov.⁸

For simplicity, one can think that all algorithms attack the following problem: given masses and natural frequencies of isotopes and a chemical compound (e.g. $C_cH_hN_nO_oS_s$), find a set of *peaks* placed at masses M_i with height/probability $\hat{P}_i$, that well approximate the full theoretical mass spectrum of that compound $\mathcal{MS}$. The direct motivation behind solving this problem, is that set $\mathcal{MS}$ contains too many configurations that, even considered together, are still improbable.

In the above setting, assigning a specific isotopologue to a peak ($\hat{M}_i, \hat{P}_i$) is not important, or down-right impossible, as that peak can very well represent a collection of isotopologues¹. In contrast, every peak generated by the (unprocessed) data-flow of **IsoSpec2** (or any other ∞ -resolution isotopic calculator, like **enviPat** or **Ecipex**) can always be traced back to a specific

¹Note that the mathematical measure theory is thus the appropriate language to globally describe all of the attempts being made here: all algorithms essentially try to establish a probability measure that is in some sense similar to the theoretical infinitely resolved mass spectrum measure. In that sense, $\mathcal{MS}$ is itself nothing else but a mixture of Dirac point measures centered at masses of isotopologues M and weighted by the isotopologue probabilities P .

isotopologue. Depending on the use case, this information may or may not be important. Let us also note here, that all of the ∞ -resolution isotopic calculators focus on finding a small yet very probable subset of the spectrum set $\mathcal{MS}$.

To approximate the masses in $\mathcal{MS}$, Mercury2 essentially replaces all of the mass differences in Eq. 4 by appropriate multiples of one mass difference Δm , or

$$\begin{aligned}\Delta m_{13\text{C}} &\approx \Delta m_{2\text{H}} \approx \Delta m_{15\text{N}} \approx \\ \Delta m_{17\text{O}} &\approx \Delta m_{33\text{S}} \approx \Delta m \\ \Delta m_{18\text{O}} &\approx \Delta m_{34\text{S}} \approx 2\Delta m \\ \Delta m_{36\text{S}} &\approx 4\Delta m,\end{aligned}\quad (6)$$

which approximates M with $\tilde{M}$ given by

$$\begin{aligned}\tilde{M} &= M_{\text{mono}} + \\ \Delta m(c_1 + h_1 + n_1 + o_1 + s_1) &+ \\ 2\Delta m(o_2 + s_2) &+ 4\Delta m s_4.\end{aligned}\quad (7)$$

Moreover, it is reasonably assumed that compounds of interest will almost always include one of the +1 neutron isotopes, e.g. ^{13}C or ^2H , so that the actual consecutive values taken by $\tilde{M}$ are as simple to represent as

$$\tilde{M} = M_{\text{mono}} + j\Delta m, \quad (8)$$

where $j \in \{0, 1, 2, \dots, L\}$, where

$$L = c(i_c - 1) + \dots + s(i_s - 1) \quad (9)$$

is the maximal number of *excess* neutrons. We define the excess neutrons for an isotopologue to be its total number of neutrons it has minus the total number of neutrons that the lightest isotopologue has. Otherwise said, its the excess with respect to what the lightest isotopologue has. Above, $i_c = 2$ is the number of stable isotopes of carbon, etc. Incidentally, there is only one such isotopologue, the heaviest one.

Calculations of M_{mono} is done using Eq. 5. Mercury2 also offers a variance-based scheme for the choice of Δm .

Consider now one atom of some element E that has i isotopes. Its isotopic distribution can be in general encoded in a multivariate gener-

ating function E

$$E = p_1 y_1^{m_1} + p_2 y_2^{m_2} + \dots + p_i y_i^{m_i}, \quad (10)$$

where p_a are the natural frequencies and the corresponding m_a masses of isotopes, and i is the overall number of isotopes of E. y_i do not have any actual physical interpretation and are used as a book-keeping device. This becomes clear upon raising E to the n^{th} power,

$$E^n = \sum_{n_1 + \dots + n_i = n} \left[\binom{n}{n_1 \dots n_i} p_1^{n_1} \dots p_i^{n_i} \times y_1^{m_1 n_1} \dots y_i^{m_i n_i} \right] \quad (11)$$

as the coefficients of the polynomial are equal to the terms of the multinomial probability function. Numbers $(n_1, \dots, n_i)$ are simply isotope counts, which we can identify with a subisotopologue. In standard chemical notation, vector $(n_1, \dots, n_i)$ corresponds to ${}^1\text{E}_{n_1} \dots {}^i\text{E}_{n_i}$.

To retrieve results for a whole molecule $\text{C}_c\text{H}_h\text{N}_n\text{O}_o\text{S}_s$ one constructs a separate polynomial for each element and multiplies them together

$$Q = C^c H^h N^n O^o S^s = \sum_{\text{iso}} P_{\text{iso}} \times \text{pol}_{\text{iso}}(y^{12\text{C}}, y^{13\text{C}}, \dots, y^{36\text{S}}), \quad (12)$$

where P_{iso} is the probability for one given isotopologue as in Eq. 2 and pol_{iso} is the corresponding polynomial of all book-keeping variables.

The trick now is to notice, that projecting variables on a lower dimensional space results in formation of clusters of isotopologues. Also, the probabilities of these clusters can be obtained from RHS of Eq. 12. For example, consider $y_a = x^{(a-1)/m_a}$. This is a nonlinear projection on a line, as all variables y_a are expressed by means of one variable only, x . Plug it in Eq. 10 to retrieve the basic building block of the BRAIN algorithm used for obtaining the probabilities,

$$\tilde{E} = p_1 + p_2 x + p_3 x^2 + \dots + p_i x^{i-1}. \quad (13)$$

The absence of masses next to p_1 corresponds to

shifting the mass axis to M_{mono} . Also, observe that consecutive powers of x differ by one. This could correspond to putting $\Delta m = 1$ in Eq. 8. Eq. 8 is what BRAIN algorithm could have used to obtain masses if it was not using a more precise algorithm, that we will return to later.

To get now the isotopic distribution of n atoms of E, or E_n , we plug $y_a = x^{(a-1)/m_a}$ in Eq. 11,

$$E^n = \sum_{n_1+\dots+n_i=n} \left[\binom{n}{n_1 \dots n_i} p_1^{n_1} p_2^{n_2} \dots p_i^{n_i} \times x^{0*n_1+1*n_2+\dots+(i-1)n_i} \right] \quad (14)$$

Observe, that the power of x is equal to the number excess neutrons² in a subisotopologue of E_n . We compactly denote subisotopologue counts $\{n_a\}_{a=1}^k$ as $\mathbf{n}$. Define the number of excess neutrons $\mathcal{N}_s$ for subisotopologue $\mathbf{n}$ as $\mathcal{N}_s(\mathbf{n}) = 0n_1 + 1n_2 + \dots + (i-1)n_i = j$. There can be multiple subisotopologues $\mathbf{n}$ of E_n that have the same number of excess neutrons. For instance, in case of O_2 both $^{17}O_2$ and $^{16}O_1^{18}O_1$ have 2 excess neutrons, as the lightest subisotopologue $^{16}O_2$ has 16 (8 per each atomic core), $^{17}O_2$ has 18 (9 per core) and $^{16}O_1^{18}O_1$ also has 18 (8 in one and 10 in another). Collect all of subisotopologues with j excess neutrons and call their combined probability q_j . Then, Eq. 14 simplifies to

$$E^n(x) = \sum_{j=0}^{(i-1)n} q_j x^j, \quad (15)$$

where

$$q_j = \sum_{\mathbf{n}: \mathcal{N}_s(\mathbf{n})=j} \binom{n}{n_1 \dots n_k} p_1^{n_1} p_2^{n_2} \dots p_{I-1}^{n_k} \quad (16)$$

is the probability of meeting a subisotopologue with j excess neutrons. The summation is performed over all subisotopologues that result in j excess neutrons. When $j = 0$, there is only one subisotopologue that contributes to q_0 - the

²So it is defined as the total number of neutrons minus the number of neutrons that the lightest subisotopologue has.

lightest monoisotopic subisotopologue. Note also that indeed the maximal number of excess neutrons across all subisotopologues $\max_{\mathbf{n}} \mathcal{N}_s$ equals to $(i-1)n$.

To get from here to $C_c H_h N_n O_o S_s$, one considers a polynomial Q equal to the product of such expressions for each element,

$$Q = C^c H^h N^n O^o S^s = \sum_{l=0}^L Q_l x^l, \quad (17)$$

where $L = \sum_e (i_e - 1)n_e$ is the number of excess neutrons that the heaviest isotopologue of $C_c H_h N_n O_o S_s$ has. To be more precise, add an extra index to subisotopologue counts, $\mathbf{n}_e$, to denote its element of origin. Also, we overwrite now the previous notation. From now on, by $\mathbf{n}$ we denote isotopologue counts, rather than subisotopologue counts. Hence $\mathbf{n} = (\mathbf{n}_e)_{e \in E}$. We can define now the number of excess neutrons for an isotopologue as the sum of excess neutrons for its subisotopologues, $\mathcal{N}(\mathbf{n}) = \sum_{e \in E} \mathcal{N}_s(\mathbf{n}_e)$. Q_l this time represents the probability of observing an isotopologue with l excess neutrons. Thus, compared to q_j , Q_l combines neutrons from different elements, of simply

$$Q_l = \sum_{\mathbf{n}: \mathcal{N}(\mathbf{n})=l} P_{\mathbf{n}}, \quad (18)$$

where each $P_{\mathbf{n}}$ is a product of multinomial distributions like in Eq. 2.

Remarkably, XVIIth century mathematicians managed to spot that coefficient Q_j can be calculated from all previous coefficients Q_a , $a < L$, and roots of polynomials C, H, N, O and S . These are the Newton-Girard formulæ, and BRAIN uses them not only to calculate the probabilities, but also, as we shall see, masses. We refer the reader to the seminal BRAIN paper⁵ to better understand how these formulæ are applied. We would like to point out, however, that procedures using roots of polynomials should be tested with respect to their numerical stability.

Mercury2 uses a different perspective on how polynomial Q can be used for calculations. In our exposition of the problem we are heavily inspired by the chapter on the Fast Fourier Trans-

form as presented in the *Introduction to Algorithms*⁹ manual and we refer the reader to this source for the overall exposition of the topic. In Mercury2 one assumes that $x = e^{2\pi it}$ and applies FFT to calculate the sequence $\{Q_l\}_l^L$. The gist of the idea is the following: polynomials can be represented either by coefficients $\{Q_l\}_l^L$ in Eq. 13, or by an equally sized set of pairs of points $\{(x_l, Q(x_l))\}_{l=0}^L$ that interpolate this polynomial. We like to call the second representation *probing*, as the polynomial is *probed* for its values at points $\{x_l\}_{l=0}^L$ to get corresponding values $\{Q(x_l)\}_{l=0}^L$. Setting x_l to complex roots of unity, one can retrieve coefficients $\{Q_l\}_l^L$ from evaluations $\{Q(x_l)\}_{l=0}^L$ in $\Theta(L \log L)$ with inverse FFT. To get the evaluations in the first place, one performs the FFT on the coefficients of polynomials C, H, N, O, S which involves only isotope frequencies. All coefficients are represented as vectors in complex space of dimension $\tilde{L}$ – the closest power of 2 that is bigger than L . Missing coefficients are set to 0. After the first step FFT, we obtain complex numbers $C(x_l), H(x_l), N(x_l), O(x_l), S(x_l)$ for each root of unity x_l . One then applies Eq. 17 to these numbers directly, which involves simple multiplications, to get numbers $\{Q(x_l)\}$. Finally, one translates $\{Q(x_l)\}$ back to coefficients $\{Q_l\}_{l=0}^L$ with inverse FFT.

We return now to how the BRAIN algorithm calculates the masses. Instead of using Eq. 8, it directly calculates the conditional masses of isotope clusters. Suppose that the j^{th} aggregated isotope peak combines isotopologues with probabilities P_{aj} and masses M_{aj} . Then, BRAIN calculates

$$\bar{M}_j = \frac{\sum_a P_{aj} M_{aj}}{\sum_a P_{aj}}, \quad (19)$$

where the individual probabilities sum up to the aggregate one $\sum_k P_{aj} = Q_j$. As Q_j are obtained using the Girard-Netwon formulæ, what remains to calculate are the denominators in the RHS of the above equation. $\bar{M}_j$ are the average masses conditional on the number of excess neutrons, which means that they minimize the probability weighted square distance to all isotopologues with j excess neutrons. If one measures error this way, they will be always closer to peaks in set $\mathcal{MS}$ than any scheme

suggested by Eq. 8, at least in terms of the weighted square distance.

To obtain $\bar{M}_j$, BRAIN makes use of a two-dimensional generating function, where another variable is introduced to store detailed masses. It modifies E from Eq. 13 into³

$$\hat{E} = p_1 K^{m_1} + p_2 K^{m_2} x + \cdots + p_i K^{m_i} x^{i-1}, \quad (20)$$

The rest proceeds as before, in that we raise the above to the number of atoms and then multiply such expressions for different elements. Recall, that we defined how to calculate the mass of an isotopologue M in Eq. 3. We shall now treat it as function of isotopologue counts. Otherwise said,

$$M_{\mathbf{n}} = \sum_{e \in E} \sum_{k=1}^{i_e} m_{ke} n_{ke} \quad (21)$$

Also, denote by $\mathbf{m} = \{n_e\}_{e \in E}$ a vector of counts of the elements, which aggregates elementwise isotope counts $\mathbf{n}$. The resulting expression can again be considered in product and sum forms,

$$\hat{Q}_{\mathbf{m}} = \prod_e \hat{E}_e^{n_e} = \sum_{l=0}^L \left(\sum_{\mathbf{n}: \mathcal{N}(\mathbf{n})=l} P_{\mathbf{n}} K^{M_{\mathbf{n}}} \right) x^l, \quad (22)$$

One differentiates $\hat{Q}_{\mathbf{n}}$ w.r.t. K and then plugs in $K = 1$. Applying it to the sum, one gets

$$\left. \frac{\partial \hat{Q}_{\mathbf{n}}}{\partial K} \right|_{K=1} = \sum_{l=0}^L \left(\sum_{\mathbf{n}: \mathcal{N}(\mathbf{n})=l} P_{\mathbf{n}} M_{\mathbf{n}} \right) x^l. \quad (23)$$

The coefficients above are precisely the denominators in Eq. 19 we search for. This motivates that we look at what we need, but does not say how to do the calculations. However, differentiating the product form of $\hat{Q}_{\mathbf{m}}$ does result in a

³Equivalently, another nonlinear substitution is performed, as $y_a^{m_a n_a} = K^{m_a} x^i$ is plugged into Eq. 10.

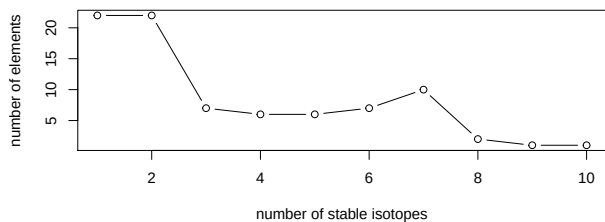


Figure S1: Number of chemical elements with a given number of stable isotopes.

tractable expression,

$$\begin{aligned} \frac{\partial \hat{Q}_{\mathbf{m}}}{\partial K} &= \frac{\partial \prod_e \hat{E}_e^{n_e}}{\partial K} = \\ \sum_e \hat{Q}_{\mathbf{m}-\mathbf{1}_e} n_e \frac{\partial \hat{E}_e}{\partial K} &\stackrel{K=1}{=} \\ \sum_e Q_{\mathbf{m}-\mathbf{1}_e} n_e \sum_{k=1}^{i_e} p_{ke} m_{ke} x^{k-1}, \end{aligned} \quad (24)$$

where by $\mathbf{1}_e$ we denote a vector with one non-zero entry at coordinate e . Otherwise said, $\hat{Q}_{\mathbf{m}-\mathbf{1}_e}$ corresponds to a molecule with one atom less of element e . Note also, that in the final line we have Q instead of $\hat{Q}$, as there is no more dependence upon K . To put it differently, Q has the same form as the one studied by BRAIN to obtain probabilities.

From there, BRAIN proceeds by applying Newton-Girard formulæ to every $Q_{\mathbf{m}-\mathbf{1}_e}$ separately, and finally combines the output by summing appropriate coefficients reweighted by the easily retrievable weights $n_e p_{ke} m_{ke}$.

The presented calculations offer more robust estimates of the positions of masses. However, because of the need to calculate roots of polynomials for the Newton-Girard formulæ to work, it remains problematic to use BRAIN algorithm in general for any element with more than 4 isotopes, as root finding is a problem that is inherently poorly conditioned numerically. Depending on the application, this might be problematic, as there are quite a few elements in nature that do have more than 4 isotopes, as seen in Figure S1. One can, in principle, find these roots only once for all elements in the Mendeleev table, but that precludes passing

in custom elemental frequencies, not to mention specifying entirely custom element with its own isotopes. Infinite resolution calculators, like *IsoSpec2*, and FFT based solutions, like *Mercury2*, are entirely free of these problems. Also, there exists a 2D FFT approach that calculates masses $\bar{M}_j$ using the same idea of one dimension x to represent aggregates, and another, along K , to represent the fine details.¹⁰ We will, however, not cover it here.

Let us mark here again, that what all of the above algorithms share is that each group of peaks with the same number of neutrons is represented not by many, but by one compound peak. They do differ slightly in the positioning of this peak. *IsoSpec* takes a different approach: it represents each group of isotopologues that have the same number of neutrons by a subgroup of isotopologues. The filtering is performed so, that altogether, the total number of peaks in the representation covers a given percentage of probability weighted isotopologues. The masses of these peaks are, in contrast to the above approximations, exact.

The clustering shown above cannot be directly used by *IsoSpec2*. However, another form of clustering can be used by *IsoSpec2*, one resulting from the Poisson approximations to isotopologue probabilities P given in Eq. 2. Namely, one can consider approximating each multinomial distribution in P by a product of independent Poisson distributions.⁸ What is being used is effectively a multivariate version of the Poisson theorem. It states, that multinomial distribution converges weakly to a product of Poisson variables.¹¹ As both distributions are discrete, it means that the probability functions of multinomial distributions converge to that of the product of Poisson distributions. For example, for O_o this means that

$$\begin{aligned} \binom{o}{o_0, o_1, o_2} p_{16O}^{o_0} p_{17O}^{o_1} p_{18O}^{o_2} &\rightarrow \\ \frac{\lambda_{17O}^{o_1}}{o_1!} e^{-\lambda_{17O}} \frac{\lambda_{18O}^{o_2}}{o_2!} e^{-\lambda_{18O}}, \end{aligned} \quad (25)$$

where it is assumed that the expected numbers of the less probable isotopes stabilize as the overall number of atoms grows to infinity.

Otherwise said, $op_{17O} \rightarrow \lambda_1$ and $op_{18O} \rightarrow \lambda_2$, where λ_1 and λ_2 are fixed numbers (the mean values of individual Poisson distributions). Although the expression looks similar, already one can spot one problem, which is that in case of the Poisson distribution numbers o_1 and o_2 are not constrained above and can take any integer values. This is relatively minor an issue but should be remembered when designing algorithms using that scheme.

On the other hand, thanks to this approximation, the binomial expressions in P from Eq. 1 can be broken and the overall probability can be written down as

$$\tilde{P} = \frac{\lambda_{13C}^{c_1}}{c_1!} \frac{\lambda_{2H}^{h_1}}{h_1!} \frac{\lambda_{15N}^{n_1}}{n_1!} \frac{\lambda_{17O}^{o_1}}{o_1!} \frac{\lambda_{33S}^{s_1}}{s_1!} e^{-\lambda_1} \times \frac{\lambda_{18O}^{o_2}}{o_2!} \frac{\lambda_{34S}^{s_2}}{s_2!} e^{-\lambda_2} \times \frac{\lambda_{36S}^{s_4}}{s_4!} e^{-\lambda_4}, \quad (26)$$

where

$$\begin{aligned} \lambda_1 &= \lambda_{13C} + \lambda_{2H} + \lambda_{15N} + \lambda_{17O} + \lambda_{33S}, \\ \lambda_2 &= \lambda_{18O} + \lambda_{34S}, \\ \lambda_4 &= \lambda_{36S}. \end{aligned} \quad (27)$$

With respect to **IsoSpec2**, this reduces the number of calculations by a constant. Clearly, this calls for another form of mass approximation, compatible with what we call *equa-transneutronic groupings*,¹² that merge counts of isotope variants by the number of extra neutrons these variants add up into the mass w.r.t. the lightest isotopologue, or simply

$$\begin{aligned} e_1 &= c_1 + h_1 + n_1 + o_1 + s_1 \\ e_2 &= o_2 + s_2 \\ e_4 &= s_4. \end{aligned} \quad (28)$$

This groupings offer an alternative scheme of marginalizing probabilities P in Eq. 2. What is more, as product of independent Poisson distributions is again a Poisson distribution,¹³ another closed formula is easily retrieved,

$$\bar{P} = \frac{\lambda_1^{e_1}}{e_1!} \frac{\lambda_2^{e_2}}{e_2!} \frac{\lambda_4^{e_4}}{e_4!} e^{-\lambda_1 - \lambda_2 - \lambda_4}. \quad (29)$$

The above expression probability defines a probability function over a rather peculiar grid, tight to the money exchange problem, as $e_1 + 2e_2 + 4e_4$ must all sum to the total number of atoms in the formula,

$$e_1 + 2e_2 + 4e_4 = c + h + n + o + s. \quad (30)$$

This time, for each e a different mass difference Δm_e has to be obtained, one for each group of isotopes (representing single additional neutrons, pairs of additional neutrons, and quadruples of additional neutrons). This can be simplest done by averaging the respective mass differences. For instance,

$$\begin{aligned} \Delta m_1 &= \left[(m_{13C} - m_{12C}) + (m_{2H} - m_{1H}) + (m_{15N} - m_{14N}) + (m_{17O} - m_{16O}) + (m_{33S} - m_{32S}) \right] / 5, \\ \Delta m_2 &= \frac{m_{18O} - m_{16O} + m_{34S} - m_{32S}}{2}, \\ \Delta m_4 &= m_{36S} - m_{32S}. \end{aligned} \quad (31)$$

Observe, that if original isotopologues of $C_cH_hN_nO_oS_s$ naturally lie in a 8-dimensional space (1D for C, H, N, 2D for O, and 3D for S), then this approximation is only 3 dimensional and still offers a peep at resolutions lower than those considered by **BRAIN** or **Mercury2**. However, that the above scheme hinges on the convergences similar to those in Eq. 25. In practice, one uses the same set of estimates of natural frequencies for all the calculations, so that numbers p_{16O} , p_{17O} , p_{18O} are fixed. Therefore, the quality of approximation varies depending on the number of considered atoms c, h, n, o, s and should deteriorate for larger molecules, as ratios $o_1 p_{17O}$ and $o_2 p_{18O}$ are not bounded. The departure of this approximation from the basic model is thus difficult to control. One might think, however, that the above approximation offers a new model of isotopic distributions, an alternative to the basic one. In that case, however, a statistical test comparing the two distributions to real data would have to be performed.

3 Order of calculations

We present the order of performing calculation by IsoSpec2 in Figure S2 by means of a standard flowchart.

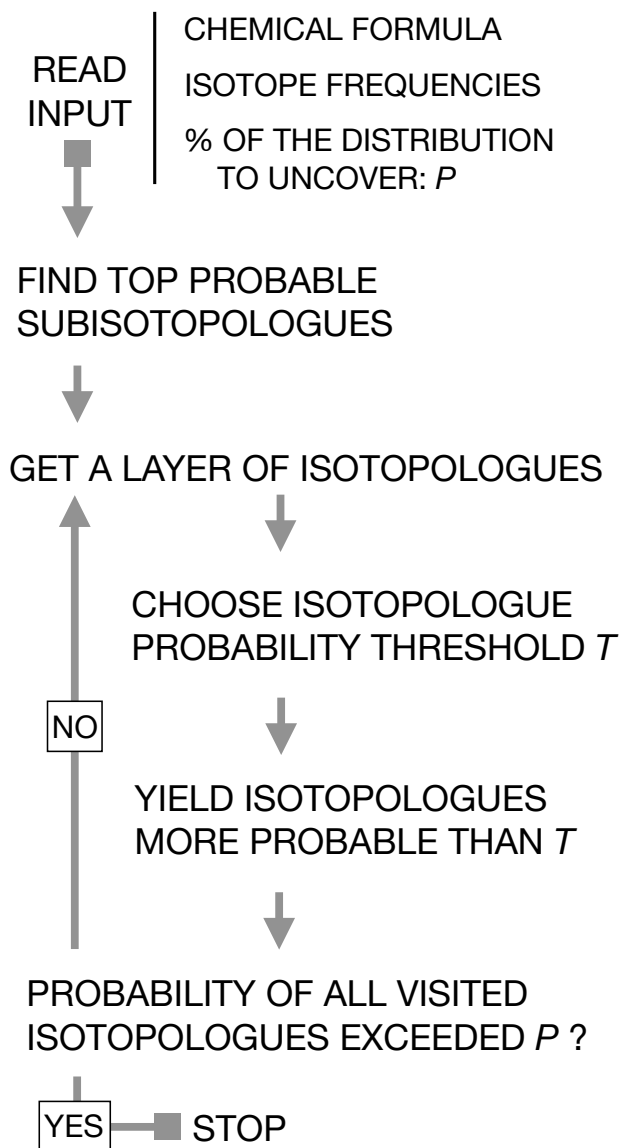


Figure S2: Order of calculations performed by the IsoSpec2 algorithm.

4 Subisotopologues

In Figure S3 we visualize the process of generating isotopologues on a dummy submolecule that consists of 25 atoms. Each atom can be in one of three isotopic variants, with probabilities equal to 60, 25, and 15%. The multi-

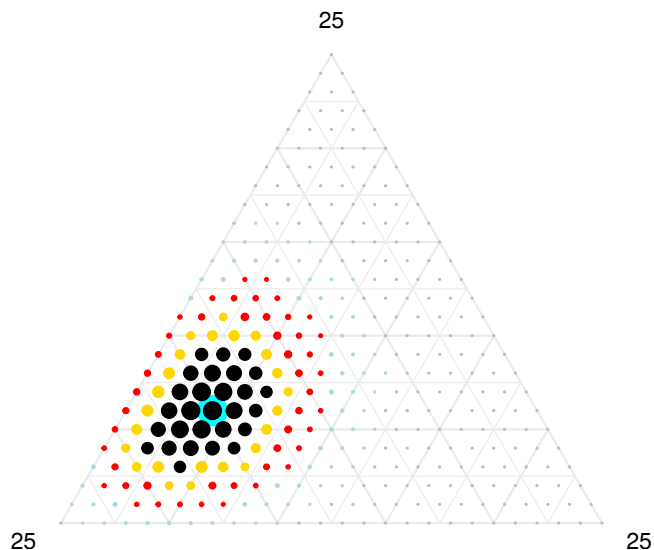


Figure S3: Layered generation of subisotopologues. Sizes are proportional to their probability. Top probable subisotopologue is highlighted in cyan.

nomial distribution is unimodal, in the sense that its probability maxima are neighbors on the simplex. This allows for finding the maximal configuration (in Figure S3 denoted by cyan halo) by a simple hill-climb algorithm. Then, consecutive layers are calculated with a sequence of submolecule-specific thresholds, $S_1, S_2, \dots$, relative to the probability of the top probable subisotopologue. Otherwise said, first layer consists of all subisotopologues higher-equal than S_1 times that probability, second layers consists of subisotopologues with probabilities in $[S_2, S_1)$, and so on. The elements of this sequence are obtained from sequence of isotopologue-probability cut-offs $T_1, T_2, \dots$, mentioned in the main manuscript and defined later on, by dividing these elements by the probability of the top probable isotopologue. The reason for this can be easily seen in Figure S4. Our primary concern is to calculate a layer of isotopologues above a given threshold. However, the border of this set (elements on the axes) correspond directly to the subisotopologues. The trace of the set of isotopologue on the axis defines the extent to which one has to move along that direction, or, how many subisotopologues to calculate.

With a first threshold in place, we perform a breadth-first-search-like descent from the top-

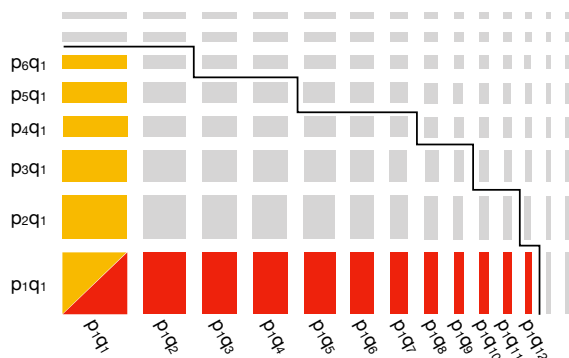


Figure S4: Finding the maximal number of subisotopologues in the *threshold* version of IsoSpec2. For each element, it is enough to follow the base axis until the threshold is reached. The black line corresponds to some optimal P -set. This set covers precisely as many subisotopologues, as there are colored in red and orange.

probable subisotopologue until all of the subisotopologues more probable than S_1 are found. To avoid double visits to the same subisotopologues, we use a hash-table. After the first layer is obtained (in Figure S3 in black), we sort the obtained subisotopologues and store them in an array. As next layers are computed (in Figure S3 in yellow and red), this table is dynamically extended to include more and more of the subisotopologues. Observe, that sorting must only be performed on layers, rather than on all of the set. This still results in an algorithm with complexity equal to $O(k \log(k))$, where k is the number of subisotopologues. As proved before,¹⁴ this does not upset the linearity of the overall algorithm, except for a particular case, when the molecule contains effectively one submolecule that has a nontrivial isotopic structure. For instance, in case of a fullerite C_{60} the notions of isotopologues coincide with that of subisotopologues. Have we sorted these configurations, our algorithm would effectively run in $O(n \log(n))$, n being the number of isotopologues. However, in that case, we simply switch the sorting off, avoiding that pitfall. This works, because in this particular case there is no need for merging any subisotopologues, which is the only operation that requires them to be ordered.

5 Speed-up due to formula order

The process of traversing isotopologues is similar to iterating consecutive integers in a decimal (or binary) numeral system. In the decimal system number 988 is *shorthand* for $9 \times 10^3 + 8 \times 10^2 + 8 \times 10^1$. To increase a number by one it is enough to change the last 8 to 9. Another unit increase (to 990) involves already the change of two coefficients. This operation, called *carry over*, is hence more expensive to carry out than a simple addition. In IsoSpec2, unlike in a typical numeral system, each element has its own set of *digits* – its subisotopologues⁴. Similar to digits, these can be totally ordered with decreasing probability⁵. For each digit, we have implemented the predecessor and the successor methods, as well as a *carry over* method. We have also noticed that ordering elements by decreasing number of subisotopologues needed to represent the final set of isotopologues has significant impact on the runtime of the algorithm. This is shown in Figure S5, where we plot how different orderings rank in terms of runtime speedups. The speedup is measured with respect to the permutation with the slowest runtime. Note that the median impact of choosing the best ordering increases the speed almost three times. Our choice of the ordering strategy is the following: sort the elements by the overall number of subisotopologues that are needed to obtain the set of all isotopologues. We observed, that this choice usually gets us on the top of the ranking shown in Figure S5. For that very reason, in Figure 3 it was more efficient to put the element with more subisotopologues on abscissa rather than on the ordinate.

The problem of finding out how many subisotopologues are necessary to construct the final set of isotopologues can be solved with the

⁴This is qualitatively similar to one pound being equal to 20 shillings, and one shilling being equal to 12 pence in the bygone currency system of the United Kingdom.

⁵In case of an unlikely event of some subisotopologues having the same probability, the order is fixed by the subisotopologue generation described in the previous publication.

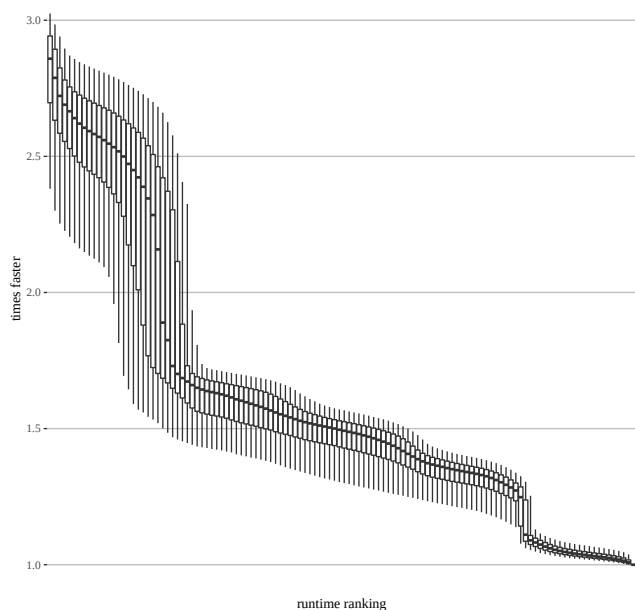


Figure S5: Effect the permutating elements in the input formula on the runtime of IsoSpec2. Each barplot corresponds to 19,767 human proteins all containing nonzero counts of each element in CHNOS. For each formula, the runtime was measured threefold 120 times (all permutations of CHNOS) and the median result is considered. Results are sorted from quickest to slowest. The boxplots show the slowest runtime divided by the result for the given permutation. This shows speed gains with respect to the slowest runtime.

threshold version of the algorithm applied to individual subisotopologues. Recall, that any algorithm starts with a hill climb to the most probable isotopologue. Incidentally, this isotopologue is composed of the most probable subisotopologues. For each subisotopologue, we *walk down* its configurations until we meet the required threshold. Figure S4 shows this procedure for a molecule with two elements. There, the top-probable isotopologue is p_1q_1 probable. We then establish the first i for which p_1q_{i+1} is below T_0 , and then the first j s.t. $p_{j+1}q_1$ below T_0 . In Figure S4, this results in $i = 12$ and $j = 6$. Therefore, subisotopologues need not be reordered and isotopologues will be generated row-wise rather than column-wise. Finding the subisotopologues is necessary for the construction of isotopologues anyway, so no calculations

are doubled. The number of subisotopologues is also (usually) limited with respect to the number of isotopologues, requiring a fairly limited memory usage. Storing them is also part of the memoization technique used to speed up their generation, as described previously.¹⁴

Following the naming conventions from the previous publication,¹⁴ let us call the outcome of the layered algorithm an optimal P -set. An optimal P -set is simply the smallest set of isotopologues with a joint probability at least P .

The problem of estimating the number of subisotopologues becomes more involved in case of the *layered* version of the algorithm, as the probability of the minimal peak in the P -set of interest is not known in advance. We can, however, estimate the absolute numbers of subisotopologues by making use of the *central limit theorem*. The argument we present below deepens (and simplifies) the reasoning we used previously¹⁴ to calculate the asymptotic runtime of the algorithm.

Consider first the task of estimating the number subisotopologues in an optimal P -set for a molecule comprised of one element⁶. Denote that number by N_{opt} . For such a molecule, the notions of a subisotopologue and isotopologue coincide. An element with n atoms and i isotopes can be fully described by a set of i integers, $(n_1, \dots, n_i)$, where $n_j \geq 0$, which is to say that we can divide all atoms into i groups corresponding to different isotopes. These sum up to n , $n_1 + \dots + n_i = n$. The optimal P -set of subisotopologues is discrete and contained within a regular simplex with edge equal to n . In Figure S6 we show this on an example of a fictitious compound made up of 100 atoms of one element with 3 isotopes. Subisotopologues within one particular optimal P -set are depicted as small dots within the orange region. This set of subisotopologues can be approximated by an ellipse (and, in general, by an ellipsoid) $\tilde{\mathcal{E}}$. Let us consider now only the first $i - 1$ coordinates of any subisotopologue, $\mathbf{n} = (n_1, \dots, n_{i-1})$. Call $\mathbf{n}$ a *reduced subisotopologue*. Since we know n , we can always retrieve

⁶The chemical impossibility of such case (except for compounds such as fullerenes) is not important for the mathematics of the problem.

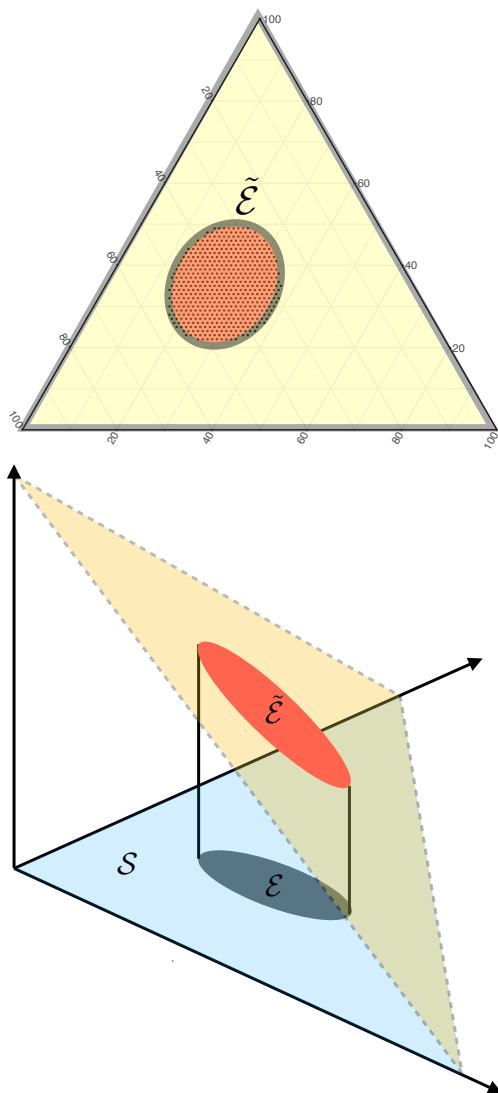


Figure S6: Top: a regular simplex with an ellipsoid $\tilde{\mathcal{E}}$ covering the optimal P -set of subisotopologues of an fictitious element with three isotopes. Below, the same simplex represented in the typical coordinate system, where each axis corresponds to counts of atoms of a given isotope. The ellipse containing approximating the optimal P -set is projected unto the plane defined by the first two isotopes (out of the three). The projection is another ellipsoid $\mathcal{E}$ contained within the (unregular) simplex $\mathcal{S}$.

the original subisotopologue from the reduced one. We illustrate that procedure in the lower part of Figure S6. The number of subisotopologues in $\tilde{\mathcal{E}}$ is the same as the number of reduced subisotopologues inside its projection, $\mathcal{E}$. Denote by N the overall number of subisotopologues on the simplex, or reduced subisotopo-

logues on the projection of the simplex. Let S denote the convex envelope of all possible reduced subisotopologues, just like in Figure S6. It comprises points

$$S = \{(x_1, \dots, x_{i-1}) \geq 0 : x_1 + \dots + x_{i-1} + x_i = n\},$$

We can now propose two schemes for estimating the total number of subisotopologues in an optimal P -set of subisotopologues⁷. The first one is straightforward: approximate it by the volume of the approximative ellipsoid $|\mathcal{E}|$, so that

$$N_{\text{opt}} \approx |\mathcal{E}|. \quad (32)$$

Another possibility is to normalize both sides to the overall dimension of the space they consider, or

$$\frac{N_{\text{opt}}}{N} \approx \frac{|\mathcal{E}|}{|\mathcal{S}|}, \quad (33)$$

where N is the total number of isotopologues and $\mathcal{S}$ is the total volume of the simplex. There is small differences between the two, that shows itself only for small numbers. Indeed,

$$N = \binom{n+i-1}{n} = \frac{(n+i-1)!}{(i-1)!n!} = \frac{\prod_{a=1}^{i-1} (n+j)}{(i-1)!} = \frac{n^{i-1}}{(i-1)!} \prod_{a=1}^{i-1} \left(1 + \frac{j}{n}\right). \quad (34)$$

What is more, $|\mathcal{S}| = n^{i-1}/(i-1)!$, a term that appears above. After reordering Eq. 33, we see that

$$N_{\text{opt}} \approx |\mathcal{E}| \frac{N}{|\mathcal{S}|} = |\mathcal{E}| \prod_{a=1}^{i-1} \left(1 + \frac{j}{n}\right). \quad (35)$$

Denote $H(n, i) = \prod_{a=1}^{i-1} \left(1 + \frac{j}{n}\right)$. For a fixed i , $\mathcal{O}(H(n, i)) = 1$, indicating that Eq. 32 and Eq. 35 are the same and the correction does not play part in the asymptotics of the problem. In our experiments, we have seen that for small optimal P -sets the above approximation predicts the correct numbers better.

⁷We define the optimal P -set of subisotopologues similarly to the optimal P -set of isotopologues as the minimal set of subisotopologues above a certain joint probability P .

The volume of an ellipsoid with radius R and the corresponding bilinear form⁸ Σ equals

$$|\mathcal{E}| = \frac{(\pi R^2)^{(i-1)/2}}{\Gamma(\frac{i-1}{2} + 1)} \sqrt{\det \Sigma}.$$

Recall, that the distribution of $\mathbf{n}$ is multinomial with probabilities equal to the natural frequencies of isotopes and the number of trials equal to the number of atoms. Otherwise said, the probability that a random reduced subisotopologue $\mathbf{N}$ equal $\mathbf{n}$ can be calculated using

$$\mathbb{P}(\mathbf{N} = \mathbf{n}) = \binom{n}{n_1, \dots, n_{i-1}, n_i} p_1^{n_1} \dots p_{i-1}^{n_{i-1}} p_i^{n_i}.$$

To avoid the cumbersome description of the multinomial distribution on the regular simplex, we project it one dimension below by considering only the first $i - 1$ coordinates of $\mathbf{n}$, which we denote by $\tilde{\mathbf{n}}$. Similarly, let $\tilde{\mathbf{p}}$ denote the first $i - 1$ probabilities in $\mathbf{p}$. As the overall number of atoms is fixed to n , the missing coordinate of $\mathbf{n}$ can be retrieved from $\tilde{\mathbf{n}}$ by subtracting the sum of its coordinates from n . Similarly, the probability term missing in $\tilde{\mathbf{p}}$ can be retrieved by subtracting the sum of the other probabilities from 1.

The mean of the distribution of $\tilde{\mathbf{N}}$ equals $n\tilde{\mathbf{p}}$ and the covariance matrix $\Sigma = n(\mathbf{d}(\tilde{\mathbf{p}}) - \tilde{\mathbf{p}}\tilde{\mathbf{p}}')$, where $\mathbf{d}(\tilde{\mathbf{p}})$ denotes a matrix with elements of vector $\tilde{\mathbf{p}}$ on the diagonal⁹. We would like to approximate the multinomial distribution with a normal one, so that the approximation has the same means and the covariance matrix. As a consequence of the *Central Limit Theorem*, that approximation is asymptotically exact as n grows. It is thus natural to choose $\mathcal{E}$ to be the ellipsoid containing P probability of the normal approximation to the random vector $\tilde{\mathbf{N}}$. Otherwise said, the above Σ and that in expression for $|\mathcal{E}|$ coincide. To calculate the determinant of Σ we use the following relation

$$\det(\mathbf{I} + \mathbf{u}\mathbf{v}') = 1 + \mathbf{u}'\mathbf{v},$$

⁸I.e. of a set $\{\mathbf{x} : \mathbf{x}'\Sigma^{-1}\mathbf{x} \leq R^2\}$ – we can neglect the exact position of the ellipsoid, as long as lies within set S . This is asymptotically always the case.

⁹And $\tilde{\mathbf{p}}'$ is the transpose of $\tilde{\mathbf{p}}$.

which follows, for instance, from the expression for a characteristic polynomial. The derivation of $\det \Sigma$ proceeds as follows:

$$\begin{aligned} \det \Sigma &= n^{i-1} \det(\mathbf{d}(\tilde{\mathbf{p}}) - \tilde{\mathbf{p}}\tilde{\mathbf{p}}') = \\ &= n^{i-1} \det \mathbf{d}(\tilde{\mathbf{p}}) \det(\mathbf{I} - \mathbf{d}(\tilde{\mathbf{p}})^{-1} \tilde{\mathbf{p}}\tilde{\mathbf{p}}') = \\ &= n^{i-1} p_1 \dots p_{i-1} \det(\mathbf{I} - \mathbf{1}\tilde{\mathbf{p}}') = \\ &= n^{i-1} p_1 \dots p_{i-1} (1 - \mathbf{1}'\tilde{\mathbf{p}}) = \\ &= n^{i-1} p_1 \dots p_{i-1} (1 - p_1 - \dots - p_{i-1}) = \\ &= n^{i-1} p_1 \dots p_{i-1} p_i, \end{aligned}$$

where $\mathbf{1}$ denotes a vector of ones. Also, we use the fact that $\mathbf{d}(\tilde{\mathbf{p}})^{-1} = \mathbf{d}(\tilde{\mathbf{p}}^{-1})$. This means, that for this special form of covariance matrix all the calculations can be performed trivially, leading to the final form of the expression for the volume of the ellipsoid $\mathcal{E}$,

$$|\mathcal{E}| = \frac{(\pi n R^2)^{(i-1)/2}}{\Gamma(\frac{i+1}{2})} \sqrt{p_1 \dots p_i}.$$

Observe that the above formula would not change had we chosen to neglect some other isotope count rather the i -th one. Otherwise said, our choice of the axis along which we projected the distribution did not play any role.

To compute the square of the radius appearing above, it is enough to note that it defines an ellipsoid that contains P probability of the normal approximation and that it follows the chi-square distribution with a certain number of degrees of freedom¹⁰, df . For a molecule composed of one type of elements, $df = i - 1$.

6 Number of isotopologues

Thanks to the simpler calculations that we obtain by projecting the multinomial distribution along one of the axis, we can further simplify the asymptotic expression for the total number of isotopologues. It amount of combining the results obtained for subisotopologues in the product space. It is a property of the normal distribution, that a product of two independent

¹⁰The degrees of freedom describe the dimension of the smallest linear space that contains all of the probability of the normal approximation.

multivariate normal distributions is again a normal distribution, with a mean that consist of concatenating the two means and a covariance matrix in a block form

$$\begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix},$$

where Σ_1 and Σ_2 are individual covariance matrices. For a general formula with $\#E$ different elements, the covariance matrix Σ would simply have $\#E$ blocks for each element on the diagonal. The determinant of that matrix would equal

$$|\Sigma| = \prod_{e \in E} |\Sigma_e| = \prod_{e \in E} \prod_{a=1}^{i_e} p_{ae},$$

where we have used the results from the previous section to express each determinant as a function of individual natural frequencies of isotopes p_{ae} .

In the product space, one should consider one bigger ellipsoid and not a product of individual lower-dimensional ellipses/ellipsoids. What does change, is the simply the calculation of the square of its radius. For one element, we have mentioned that one needs to consider the P^{th} quantile of the χ^2 distribution with $i_e - 1$ degrees of freedom. Now, we have to simply change that last number to $k = \sum_{e \in E} (i_e - 1)$, which is the dimension of the new ellipsoid. Putting it all together, the asymptotic expression for the total number of isotopologues N_{opt} is

$$N_{\text{opt}} \approx \frac{(\pi q_{\chi^2(k)}(P))^{k/2}}{\Gamma(\frac{k}{2} + 1)} \sqrt{\prod_e \prod_a p_{ae}} \times \prod_e H(n_e, i_e) \sqrt{\prod_e n_e^{i_e-1}} \quad (36)$$

Beyond being useful for the study of the asymptotic behaviour of our algorithm, the new estimate provides qualitative insights into what ultimately controls the behaviour of the problem of finding the optimal P -set. For instance, we directly see, that the number of configurations will depend on the natural frequencies of elements. Observe, that $\prod_{a=1}^{i_e} p_{ae}$ (for which also $\sum_{a=1}^{i_e} p_{ae} = 1$) is maximal for uniform dis-

tribution on isotopes. The more one isotope dominates over the others, the lesser will be the number of isotopologues in the resulting optimal P -set. Note, that the above result does not merely describe the asymptotics. Indeed, our choice of approximation is precise in terms of the mean and covariance matrix. Using multivariate Chebyshev inequalities, one could check, that the overall number of isotopologues within an ellipsoid of some size would also directly depend upon the products of natural frequencies of isotopes. Even if the shape of the ellipsoid would not perfectly match that of the optimal P -set of the product of multinomial distribution, it would still describe the behaviour of sets of isotopologues not far off from the mode. Otherwise said, as we can definitely always cover an optimal P -set by some ellipsoid, and the ellipsoid grows or gets smaller depending on the natural frequencies p_{ae} , then so will the real optimal P -set. This way, one can at least describe its behaviour qualitatively.

Finally, note that Eq. 36 is valid only for elements with more than one isotope. Elements with only one isotope do not modify the complexity of any of the IsoSpec algorithms. Their presence only modifies the reported masses, and is also reflected in the ultimately reported isotope counts. The approximation is also valid only for a fixed $P < 1$. When $P = 1$, we have to enumerate all of the isotopologues and there are N of them. In that only case, does the complexity grow to $\mathcal{O}(\prod_{e \in E} n_e^{i_e-1})$. To be precise, it equals $\prod_{e \in E} \frac{n_e^{i_e-1}}{(i_e-1)!}$, because we simply have to consider a product of expressions describing the total number of subisotopologues for each element.

7 Real Punnett tiles

In the Methods section of the main manuscript we have been using a dummy molecule C_{100}O_6 with modified probabilities to visualize the ideas behind the calculations. In Figure S7 we show how such tiling looks for calculations based on real natural frequencies, for a very long polymer $\text{C}_{100000}\text{H}_{200002}$.

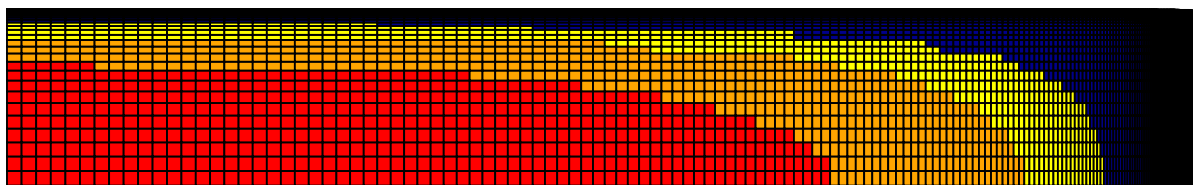


Figure S7: Punnett tiles for $C_{100000}H_{200002}$ with isotope frequencies as they occur in Nature. In colors, we show three layers of 500 isotopologues (red, orange, yellow), and in blue the remainder. These isotopologues cover 99.9% of the theoretical mass spectrum. Tiles are scaled, so that the mode isotopologue (bottom-left corner) is a square.

8 Simulations

IsoSpec2 offers additionally a fast way to simulate mass spectrum of a given substance. What we mean by that, is that for a given chemical formula we are able to calculate randomly sampled mass spectra, following the acknowledged multinomial model.

For example, for a chemical formula C_cH_h a random isotopologue consists of a formula $^{12}C_{c_0}^{13}C_{c_1}^1H_{h_1}^2H_{h_2}$ that would reappear with relative frequency of $\binom{c}{c_0, c_1} p_{^{12}C}^{c_0} p_{^{13}C}^{c_1} \binom{h}{h_0, h_1} p_{^1H}^{h_0} p_{^2H}^{h_1}$ upon repeated random drawing. Each of such randomly drawn isotopologues can be readily represented as a mass spectrum by simply calculating its mass. Upon repeating the procedure multiple times, one would obtain many randomly sampled peaks. This is, however, a highly inefficient way of performing calculations. Instead, we note that the same result can be obtained by drawing random peaks *en mass*, i.e. drawing a whole random spectrum. To do it, we note that probabilities $\binom{c}{c_0, c_1} p_{^{12}C}^{c_0} p_{^{13}C}^{c_1} \binom{h}{h_0, h_1} p_{^1H}^{h_0} p_{^2H}^{h_1}$ when multiplied by the number of molecules we want to simulate, say N , are the expected spectrum of the scheme presented above. What is more, drawing a whole spectrum corresponds to another multinomial scheme with N trials and precisely the above probabilities (incidentally, the output of the **IsoSpec** algorithm). The standard way of

efficiently drawing random samples from the multinomial distribution is known as the alias method, originally proposed by A.J. Walker.¹⁵ Unfortunately, this method requires all the frequencies to be known in advance so that they could be reordered into a 2D data structure. This makes it impractical to use in our case, as **IsoSpec** algorithm avoids calculating full isotopic distributions, concentrating only on the smallest sets with a given probability. Instead, we make use of an alternative, but equally efficient *on flight* algorithm.¹⁶

9 Complexity

Similarly to **IsoSpec1**, the overall number of operations remains linear in the size of the output. This is because the algorithm visits the same isotopologues mostly the same number of times. Multiply visited isotopologues belong to lower dimensional sets and can be safely neglected. What is more, the algorithms (except for the small subisotopologues that need to be ordered) have been freed from the use of any queue data structures for storing isotopologues intermediate to the computation. What also results, is that the generator versions of the algorithms can operate at minimal space requirements.

10 Computer specs

Our test computer specifications:

- Intel(R) Xeon(R) CPU E5-2620 v2 @ 2.10GHz (24 logical cores total)
- 64 GB RAM.

Operating system:

- Linux Gentoo x86_64 stable kernel v. 5.3.6
- System compiled with GCC 8.3.0-r2 with system-wide cflags:

– -O3 -pipe -march=native -mtune=native

- Python 3.6.9
- R version 3.4.1 (2017-06-30)
- enviPat 2.4

IsoSpec2 was compiled with GCC 8.3.0-r2 compiler with cflags:

- -O3 -march=native -mtune=native -std=c++17

11 How to run IsoSpec2?

Python. You can install the module with
`pip install IsoSpecPy`

To calculate all peaks more probable than 0.01% for $C_{254}H_{377}N_{65}O_{75}S_6$ run:

```
import IsoSpecPy as iso
```

```
spectrum_thr = iso.IsoThreshold(  
    formula="C254H377N65O75S6",  
    threshold=0.0001)
```

```
print(list(spectrum_thr.masses))  
print(list(spectrum_thr.probs))  
# list conversion is not needed  
print(spectrum_thr.masses[0])
```

One can also directly iterate the outcomes:

```
for mass, prob in spectrum_thr:  
    print(mass, prob)
```

To find the optimal 99.9%-set, run:

```
spectrum_lay = iso.IsoTotalProb(  
    formula="C254H377N65O75S6",  
    prob_to_cover=0.999)  
list(spectrum_lay.masses)  
list(spectrum_lay.probs)  
for mass, prob in spectrum_lay:  
    print(mass, prob)
```

To only generate configurations (without storing them) above a given probability threshold, run:

```
for m, p in iso.IsoThresholdGenerator(  
    formula="C254H377N65O75S6",  
    threshold=minimal_peak_height):  
    print(m, p)
```

R. To install, run:

```
install.packages('IsoSpecR')
```

To get isotopologues above a given minimal probability and to show the isotopologues, run:

```
library('IsoSpecR')  
mol = c(C=254,H=377,N=65,O=75,S=6)  
X = IsoSpecify(molecule=mol,  
    stopCondition=0.0001,  
    algo=3,  
    showCounts=True)  
  
print(X)
```

to find the optimal 99.9%-set, run:

```
Y = IsoSpecify(molecule=mol,  
    stopCondition=0.999,  
    algo=1,  
    showCounts=True)  
  
print(Y)
```

As mentioned in the manuscript, in R it makes sense not to return isotopic counts, if they are not of interest. Doing so massively speeds up calculations.

C++. Examples of running the code in C++ are available for inspection in the `git` repository.

References

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